

# The One Percent Per Week TQQQ Trading Strategy:

## The Expected Unexpected

by: Guy R. Fleury

In this article, I will look for inconsistencies in my published version of the TQQQ trading strategy's application. It will be a relevant and simple description of what is there, leading to what may be an unexpected conclusion.

Over the past few articles, I performed tens of thousands of 15-year simulations on **The One Percent Per Week** (OPPW) strategy using TQQQ's WL8 simulation metrics. For starters, no one would think of continuously putting all their trading account into a single stock, but here, I am proposing just that and using a single 3x-leveraged ETF.

It can make quite a difference to one's outlook. As we know, TQQQ tracks the QQQ ETF and wants to generate 3x QQQ's daily return. It also means more volatility, larger drawdowns, and uncertainty but a higher long-term expected return. See the [TQQQ](#) website page for its compounded return since inception in 2010.<sup>1</sup>

Again, as said in previous articles,

*I cannot predict the price move for tomorrow or next week. However, over the long term, I can observe the market's long-term trend, which, on average, has been to the upside for decades.*

Therefore, from those decades of market data, we can take averages of our trading procedures' average trading behavior and evaluate their long-term potential impact.

In my previous article, [The One Percent Per Week TQQQ Trading Strategy: MAKING IMPROVEMENTS II](#), I looked at the impact a slight change in attitude might have over the long term. Changing the built-in break-even stop-loss by -0.25% had a considerable weight on overall results. It was like instead of requesting 100% of the entry price to exit, it would now request only 99.75% to exit the trade.

Here, I want to analyze the results of the simulations and explore what the strategy could do over the coming years.

Already with Figure #2 in the above-cited article (reproduced below as Figure #1), we had a rising cumulative CAGR over the years, way above what could be considered as the market's long-term average. Within 5 years, the strategy's CAGR climbed above 60% and continued to rise. It kept on increasing over those 20 years under consideration.

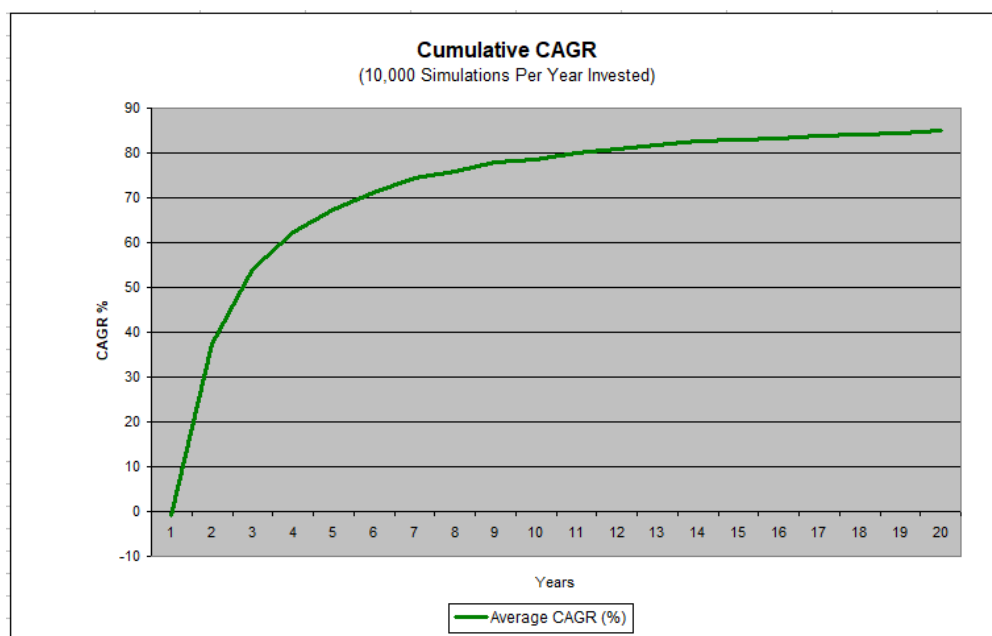
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<sup>1</sup> The average rate of return for TQQQ since February 2010 has been over 40% compounded.

The portfolio metrics used for the simulations were obtained from the 15-year simulation results from **The One Percent Per Week (OPPW)** strategy. Those are long-term averages taken from the WL8 simulation run. It is a point that needs to be emphasized. It took 780 trades to obtain those metric averages; in the future, those metrics should not change much over an added few months.

As shown in Figure #1, at the 15-year level, the CAGR exceeds 80%. So, the question becomes: how come? The Wealth-Lab simulation portfolio metrics did not generate that high a CAGR, as illustrated by Figure #4 in the above-cited article (reproduced below as Figure #2).

**Figure #1: 10,000 Simulations. Average Compounded Returns Per Year.**



([Click here to enlarge](#))

There is a need for an explanation, explicit and sustainable. Figure #1 illustrates the point that the long term matters since the compounded rate of return is increasing with time.

Figure #2, with its first improvement, had a 56.98% CAGR. In contrast, Figure #1 implies a compounded return above 80% over those same 15 years based on the average of the 10,000 Monte Carlo simulated portfolios.

Those simulations were all based on the WL8 15-year average metric results. Long-term averages that took 15 years (780 trades) to achieve.

The program code for these simulations can be found in Figure #4 of my paper: **[The One Percent Per Week TQQQ Trading Strategy: MY EQUATION.](#)**

The equation using the WL8 simulation metrics from Figure #2 had a hit rate of 50.95%. It comes out with those simulation metrics as:

$$F(t) = 0.5144 \cdot \$100,000 \cdot (1 + 0.046)^{403} \cdot (1 - 0.0270)^{388} = \$93,401,070 \quad (1)$$

which is close enough to the simulation results. The rounding on the third decimal for the average percent win and loss per trade is the main reason for the outcome difference between what the equation provides and what the simulation gave (see Figure #2).

**Figure #2: Wealth-Lab Portfolio Simulation. April 8, 2025. First Improvement.**

| Metrics Report                    |               | Equity Curve    |  |
|-----------------------------------|---------------|-----------------|--|
| Select ScoreCard: Basic ScoreCard |               |                 |  |
|                                   | Strategy      | Benchmark (Q... |  |
| Summary                           |               |                 |  |
| Starting Capital                  | 100,000.00    | 100,000.00      |  |
| Profit                            | 93,141,786.04 | 954,236.65      |  |
| Profit %                          | 93,141.79%    | 954.24%         |  |
| Profit Per Bar                    | 15.66         | 12.51           |  |
| APR                               | 56.98%        | 16.81%          |  |
| Std Dev of Annual Ret...          | 154.82%       | 21.65%          |  |
| Exposure                          | 51.44%        | 99.99%          |  |
| Maximum Exposure                  | 99.91%        | 100.00%         |  |
| EAR                               | 110.78%       | 16.81%          |  |
| Alpha (α)                         | 35.63         | -               |  |
| Beta (β)                          | 1.30          | -               |  |
| Sharpe Ratio                      | 1.25          | 0.90            |  |
| Sortino Ratio                     | 2.13          | 1.49            |  |
| WL Score                          | 50.01         | 10.82           |  |
| Slope of Equity Curve             | 18,768.09     | 261.82          |  |
| Interest, Commission...           |               |                 |  |
| Commission Paid                   | 0.00          | 0.00            |  |
| Cash Interest Received            | 0.00          | 0.00            |  |
| Margin Interest Paid              | -0.00         | -0.00           |  |
| Maximum Margin Used               | 1.00          | 1.00            |  |
| Dividends Received                | 0.00          | 0.00            |  |
| Total Currency Adj                | 0.00          | 0.00            |  |

| Metrics Report                    |                | Equity Curve    |  |
|-----------------------------------|----------------|-----------------|--|
| Select ScoreCard: Basic ScoreCard |                |                 |  |
|                                   | Strategy       | Benchmark (Q... |  |
| Positions                         |                |                 |  |
| Position Count                    | 791            | 1               |  |
| Avg Profit                        | 117,751.94     | 954,236.65      |  |
| Avg Profit %                      | 1.02%          | 954.51%         |  |
| Profit Factor                     | 1.45           | -               |  |
| Payoff Ratio                      | 1.71           | -               |  |
| Avg Bars Held                     | 3.27           | 3,815.00        |  |
| Avg Trades Per Month              | 8.64           | 0.01            |  |
| Avg Bars Held as % of...          | 0.09           | 99.97           |  |
| Largest Bars Held as %...         | 0.13           | 99.97           |  |
| NSF Position Count                | 0              | 0               |  |
| NSF Ratio                         | 0.00           | 0.00            |  |
| Drawdown                          |                |                 |  |
| Max Drawdown                      | -35,133,550.71 | -333,824.80     |  |
| Max Drawdown Date                 | 3/28/2025      | 12/28/2022      |  |
| Max Drawdown %                    | -54.86%        | -35.62%         |  |
| Max Drawdown % Date               | 7/6/2010       | 12/28/2022      |  |
| Recovery Factor                   | 2.65           | 2.86            |  |
| Profitable Positions              |                |                 |  |
| Count                             | 403            | 1               |  |
| Max Consecutive                   | 9              | 1               |  |
| % Profitable                      | 50.95%         | 100.00%         |  |
| Avg Profit                        | 739,407.65     | 954,236.65      |  |
| Avg Profit %                      | 4.60%          | 954.51%         |  |
| Average Bars Held                 | 3.00           | 3,815.00        |  |
| Unprofitable Positions            |                |                 |  |
| Count                             | 388            | 0               |  |
| Max Consecutive                   | 15             | 0               |  |
| % Unprofitable                    | 49.05%         | 0.00%           |  |
| Avg Loss                          | -527,936.85    | -               |  |
| Avg Loss %                        | -2.69%         | -               |  |
| Avg Bars Held                     | 3.54           | -               |  |

([Click here to enlarge](#))

If I have the long-term market average at about 10% over decades, should I not expect to have about the same for the next few decades?

If I have over two centuries of market data where that 10% average has prevailed, I could not advance that over the next few decades; I should have 20% or 30%

compounded return using the same average investment techniques used in the prior decades.

The first thing we brought to the table for increasing long-term returns was using TQQQ, a 3x-leveraged ETF. Therefore, we should expect about 3 times the average market return.

It would be an earned reward for accepting more volatility as long as we knew we would not lose the long-term game.

But here, we have tens of thousands of simulations where the vast majority exceeded market averages and even a high-performing strategy such as **The One Percent Per Week**.

What would be the reason? I need one.

In the Wealth-Lab 8 simulation, we have a single TQQQ price series. It is always the same path for the thousands of simulations made. You will have a different outcome if you change your trading procedures. The difference in outcome is not because TQQQ's price series has changed but because you modified your entries and exit points, generating different profit and loss series. The change need not be big, as demonstrated in my prior article, where only a -0.25% stop-loss upped the portfolio's performance by \$15 million.<sup>2</sup>

We used the portfolio metrics of equation (1) to perform those thousands of simulations. We know that at year 15, they should tend to the same values as in equation (1). Therefore, we should get the same average results as presented in Figure #2. However, we got much more in the Monte Carlo simulations using the same portfolio metrics.

The following chart shows two normal distributions with a mean of zero, one with a one-sigma and the other with a three-sigma spread. With TQQQ, what was one sigma (in red) is now spread out over 3 sigma (in cyan). We could consider the red as QQQ's volatility and view the cyan as TQQQ's volatility. TQQQ will swing a lot more, and its average simulation result will spread over a larger return space.

The TQQQ average daily percent change is a QQQ 3x-leveraged scenario.

Playing TQQQ or QQQ is a matter of choice. However, you cannot expect the same outcome.

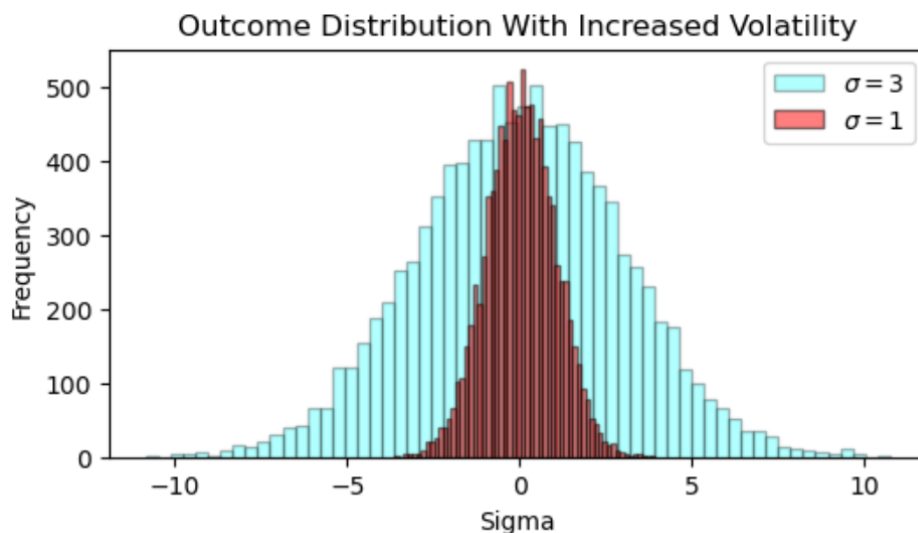
Both ETFs will be there 10, 20 years from now, and even longer. That assumption is simple: it rests on the market's evolution. All nations can grow and prosper. We

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<sup>2</sup> See: [The One Percent Per Week TQQQ Trading Strategy: MAKING IMPROVEMENTS II](#).

might be in a chaotic world, but we have and will continue to improve most people's living conditions decade after decade. Mind you, there is still much work to do to improve living conditions for all. There also lies hope for a future prosperity for all.

**Figure #3: TQQQ's Return Distribution With Increased Volatility.**



[\(Click here to enlarge\)](#)

QQQ is a good example of what we can do. For instance, you could buy 20 shares of QQQ every year and be a millionaire within 20 years.

We know QQQ is composed of 100 of the highest-valued stocks in America. The top 10 stocks are easily recognizable; we have AAPL, MSFT, NVDA, AMZN, AVGO, META, NFLX, COST, TSLA, and GOOGL. All of them will continue to prosper for years to come. How long they will stay as the top 10 richest is another matter. Whatever happens, if a stock in the top 10 is replaced, it will be by some other stock that has even better long-term prospects. Even those ten stocks were young and small some 20 years ago (most not on that list yet).

It is like justifying that QQQ will continue, on average, to trade upward over the next few decades. Therefore, you could buy and hold QQQ for many years to come. It is the same as buying a long-term and above-average market return.

Based on Figure #2, holding QQQ would have generated a 16.81% CAGR over the 15+-year testing period. It would have transformed our \$100k into \$954,236. It is much better than having used SPY over the same period. SPY would have generated an 11.18% CAGR and returned \$398,771.

You got better results just for choosing QQQ over SPY. Not only that, you already knew, 15 years ago, which of the two would outperform the other, as if it was a no-brainer, even at the time. One decision would have made you a millionaire.

Nonetheless, QQQ and SPY's returns since 2010 have been eclipsed by TQQQ's performance. It would have taken the same initial \$100k and generated \$93,141,786 over the same 15 years.<sup>3</sup>

That too, was a question of choice since TQQQ's outcome was, from the start, expected to come close to 3 times QQQ's return, or having something near a 50% CAGR level. That is also what is implied by Figure #2 above.

From Figure #2, we have an average profit per trade of 1.02%. Even at week 741, we had a 0.97% average return per trade.<sup>4</sup> Adding 50 weeks to those results and slightly improving the strategy raised the average return to this 1.02% profit per trade. The trading strategy did not break down or even slowed down.

## THE RANDOMNESS BENEATH

The changes you can make to this strategy should aim to impact equation (1). The rest would only be cosmetics. You can do cosmetics at any time for any reason, but it might not impact the final results.

The problem with TQQQ is that we have only one price series. Whatever action you take, your trading will use that price series. You are limited to trade segmentation. You can break down the time series into as many segments as you want, but they all will be one-to-one relationships to the TQQQ price series. The TQQQ price series can not change its past trajectory.

There are no grounds to claim survivorship biases.

TQQQ resets every day to track QQQ as it is. It was not possible in the past for TQQQ to be the outcome of some survivorship bias, just as it will not be able to see any survivorship bias in the future. It is not just some fundamental financial statement but just what it is.

TQQQ will continue to track QQQ, whatever its future composition. The real question should be: will TQQQ maintain its 3x-leveraged stance? One thing is sure: over the past 15 years, it did.

TQQQ deals only with survivors, the best-performing stocks. That is not survivorship bias; it is following, day by day, the best ones out there.

If you did a Monte Carlo simulation without replacements using past returns of executed trades, this strategy would give the same final answer every time. The order in which the trades occurred has no consequence on the final results.

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<sup>3</sup> See Figure #2.

<sup>4</sup> See the first article of the series: [The One Percent a Week Stock Trading Program - Part I](#).

We have a product function at play in this trading strategy:  $\prod_1^N (1 + r_i)$ . If we shuffle all the  $r_i$  around, the value of the expression will not change.

Shuffling the trade returns in this strategy will have no effect. It is even said so in equation (1) above. All this shuffling could do is change the path to reach the outcome, but the result would always be the same.

So, there is no value in making that kind of Monte Carlo simulation. It could only tell you the strategy could have a worse drawdown. The more you would do simulations, the more likely you would be able to find an even larger drawdown. And you still would not know the probability of such an event. The larger drawdowns are often the result of random events (Black Swans).

The worst drawdown for the **One Percent Per Week** strategy occurred during the May 2010 Flash Crash (-54.86%). There is nothing you could have done to bypass that drawdown except if your portfolio was in cash that day.

The worst kind of Monte Carlo testing is to have the return distribution randomly generated without using past returns but based on using randomly generated market returns facsimiles. Introducing completely new paths independent of the achieved WL8 simulation.

To make the case, consider flipping a fair coin 10,000 times. Its distribution would give the red bell-shaped curve in Figure #3. We are more than familiar with normal distributions. We know the odds of the coin flip and can simulate coin flips ad infinitum. But, it would not change the shape of the distribution. The variance would still be:  $\sigma^2 = n \cdot p \cdot q$ , and the standard deviation  $\sigma = \sqrt{n \cdot p \cdot q}$ , no matter the millions of trials.

In the WL8 simulation, we had a 50.95% hit rate (see Figure #2, profitable positions section). That is very close to the 50/50 expected value of a fair coin toss simulation. It is so close that randomness testing could not negate the no-randomness hypothesis.

That is not good news for a trading strategy having near 50/50 odds.

Equation (1) has already separated the product function  $\prod_1^N (1 + r_i)$  into its positive and negative average returns:  $(1 + 0.046)^{403} \cdot (1 - 0.0270)^{388}$ . Those numbers come directly from the WL8 simulation metrics. They were not invented numbers; they were just the result of the simulation using the OPPW strategy. Any change to the trading procedure can impact the strategy's outcome.

Figure #3 above is an expression of this randomness. You could view it as the outcome of 10,000 simulations of 10,000 coin flips. The outcome distribution would look like the red bell-shaped curve with a sigma of 1.0 and a mean of zero. We

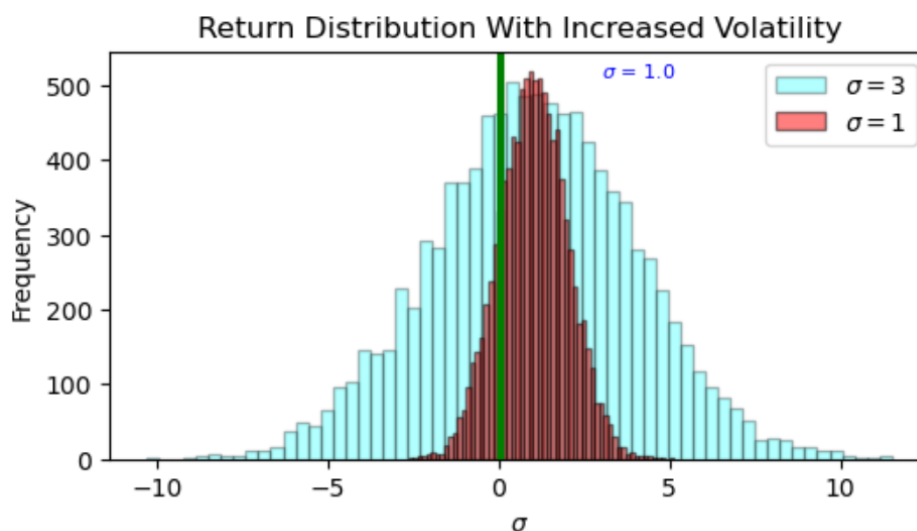
could view Figure #3 as the QQQ return distribution (in red) compared to the TQQQ distribution (in cyan).

However, returns do not follow a strict normal distribution in the market. There is an upward bias. It could be expressed as a 51/49 edge, thereby moving the mean to something like +1% instead of zero.

Figure #4 shows similar normal distributions as in Figure #3 but with the mean increased from zero to one. It shows the inserted positive bias in the distribution of returns. We can easily determine that more than half of the returns will be positive. Both curves operate under the same principle. The TQQQ strategy is 3x the level QQQ could achieve.

With Figure #3 as a base, half the returns would be above the mean due to the normal distribution. They would be the mirror image of the half below the mean. And just as in equation (1), we could find the average of the positive and negative returns.

**Figure #4: Return Distribution With Increased Mean From Zero To One.**



([Click here to enlarge](#))

We should also expect a 50/50 distribution of returns, as depicted in Figure #3. The positive and negative returns would have to be the mirror image of each other, having something like  $|\bar{r}| = 0.05$ . Using (1) with these numbers would give:

$$F(t) = 0.50 \cdot \$100,000 \cdot (1 + 0.05)^{390} \cdot (1 - 0.05)^{390} = \$18,836 \quad (2)$$

Your portfolio would not be growing but shrinking.

Your \$100k portfolio would drop to \$18,836 over those 15 years. Raising to a thousand trades under the same trading conditions would make the problem worse:



$F(t) = 0.50 \cdot \$100,000 \cdot (1 + 0.05)^{500} \cdot (1 - 0.05)^{500} = \$14,302$ . The more you would play the game, the more you would lose. If the average percent win or loss were higher, you would lose your money even faster as the number of trades increased.

If you were dealing with a normal distribution in the market, you would lose in the long term. Making playing that game irrelevant and worthless.

Therefore, we should be happy that the market does not follow a normal distribution. It is more of the Pareto type with higher kurtosis, skewness, slight upward long-term trend, and fat tails. It is still unpredictable in the short term. However, it has a long-lasting upward bias (Figure #4). That is what we can take advantage of, and easily.

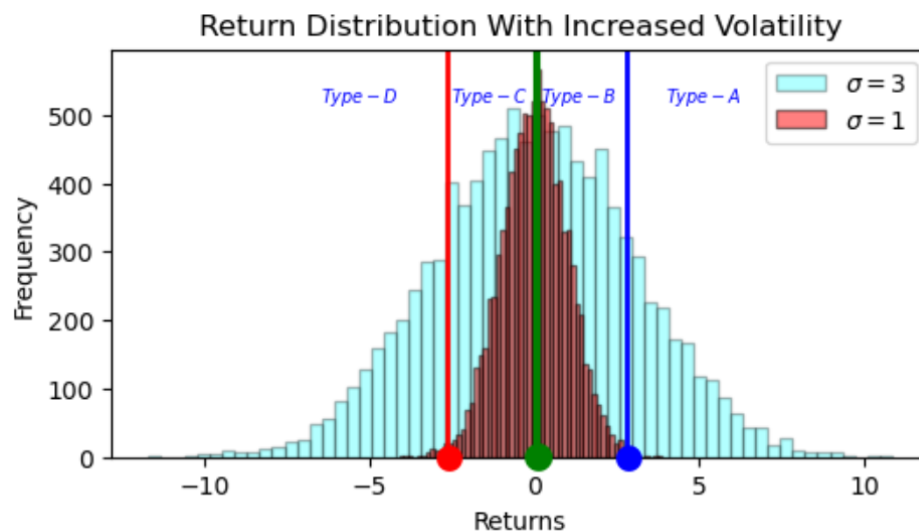
My **One Percent Per Week** strategy has an added structure in that we have broken the achieved bell-shaped curve into four trade types: two positive and two negative. If we assumed, for instance, a mirror image as in Figure #3, we could average the average return for each type.

$$F(t) = 0.50 \cdot \$100,000 \cdot (1 + 0.08)^{195} \cdot (1 + 0.03)^{195} \cdot (1 - 0.03)^{195} \cdot (1 - 0.08)^{195} = \$11,994 \quad (3)$$

That would be equivalent to a -13.18% CAGR. Again, this makes the case that the market is not following a normal distribution.

We cannot consider the stock market as totally random. Otherwise, it would have destroyed most long-term trading portfolios.

**Figure #5: Return Distribution - Mean Zero And Trade Types.**



([Click here to enlarge](#))

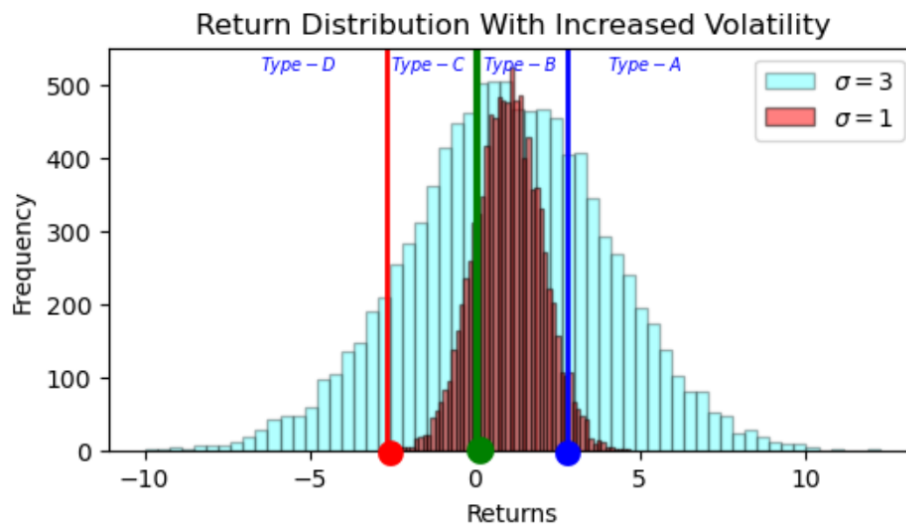
A visualization of equation (3) would give something like Figure #5 with the distribution of trade types.

But that is not what we get. Equation (3) could apply only if the market or stock returns were following a normal distribution, which it is not.

There would be a significant change if the means increased to 1%.

We can visually observe the differences in the distribution in Figure #6. There are more Type-A and Type-B trades. At the same time, there are fewer Type-D and Type-C trades. Note that the trade type separation would vary based on the simulation. Since the total number of trades is constant, the increase in Type-A trades has to come from the other types.

**Figure #6: Return Distribution - Mean 1% And Trade Types.**



[\(Click here to enlarge\)](#)

We could rewrite equation (3) to account for the trade type shift.

$$F(t) = 0.50 \cdot 100,000 \cdot (1+0.085)^{205} \cdot (1+0.031)^{205} \cdot (1-0.029)^{185} \cdot (1-0.075)^{185} = 1,126,862 \quad (4)$$

That is a great improvement. You did not request that the market have an upward bias over time; you only observed that it was built-in and had been part of the average market return for over two centuries.

But the picture is even better than that.

With my trading procedures for the **One Percent Per Week**, I changed the trade balance and the profit and loss distribution. It was not a major request. I set one of the trading rules as a break-even policy if the price fell below the entry price at the close of any trading day. In a way, chickening out on any sign of weakness.

The new Type-C procedure has quite an impact on the overall results. To make the point, equation (4) was modified to account for the new Type-C zero average

outcome.

$$F(t) = 0.50 \cdot 100,000 \cdot (1+0.085)^{205} \cdot (1+0.031)^{205} \cdot (1-0.0)^{185} \cdot (1-0.075)^{185} = 260,803,414 \quad (5)$$

That is now a significant difference in outlook and outcome. You opted to be more cautious with your Type-C trade exit on the slightest indication of weakness. Even a one-penny drop below the entry price would trigger a Type-C trade. And yet, anyone would readily accept the new long-term outcome. From equation (5), it would give you a 68.95% CAGR.

Starting with \$200k instead of \$100k would double the outcome to \$521,606,827. It is a choice you have to make if you can.

The actual future distribution of trade types will be different and unpredictable.

However, after 15 years, the number of trades will be 780. You could make the distribution you want, knowing that the sum of all the trade types will total 780. And since the market offers a 51% hit rate, you should expect 398 wins to 382 losses, of which some 185 will not impact the total by their presence (Type-C trades), but needed to make the strategy thrive.

The above explains why my version of the **One Percent Per Week** wins the game. Applying simple trading procedures and not needing outside help, whether fundamental data, technical indicators, or pseudo-data, is nothing extraordinary.

You do not have to rely on moon phases, astrology, market gurus, talking heads, or sentiment analysis. You are playing the math of the game.

My version of the **One Percent Per Week** is not dealing with opinions. It just games the system to its advantage. And its biggest gains come from its chicken stance to price variations. Surprisingly, or not surprisingly.

## THE UNEXPECTED

The future remains the future and may still be considered elusive.

We have a hard time making estimates that actually can come true. Just look at forecasts made 10, 20, or 50 ago. For instance, in the 1970s, "experts" told us we were entering a new ice age within 10 to 20 years. How did that turn out?

I used the 15-year WL8 simulation metrics as the foundation for making simulation to arrive at the same 15-year portfolio metrics. Therefore, my simulations should tend to these 15-year averages.

At least, that is what I should have expected. If the market, on average, has had a 10% average return over the past 100 years, I should expect to get about the same

results over the next 30+ years. However, on that same outlook, as showcased in my last articles, my simulations exceeded their expected long-term portfolio values.

We still need a reasonable explanation for this phenomenon.

Equation (5) makes its point. The Type-C trades generate no profits and make you lose time since those weeks hold no profits. Yet, they are the reason the trading strategy wins.

Using the WL8 simulation metrics in a Monte Carlo scenario with randomly generated trade types, we get more, on average, than what should have been expected. On the 15<sup>th</sup> year, we should expect the portfolio metrics to reach the same level as those in the WL8 simulation. However, using those portfolio metrics and doing 10,000 simulations based on those trade statistics, we get something like the following:

### Figure #7: WL8 Simulation Using OPPW Portfolio Metrics.

```
# of Sims: 10,000 over 15 years using randomly selected trades based on WL8 sim metrics.
Total: $ 406,772,756 Sim #: 1 (A: 134, B: 257, C: 239, D: 150) CAGR: 74.03 %
Total: $ 1,443,949,055 Sim #: 1,000 (A: 144, B: 269, C: 219, D: 148) CAGR: 89.37 %
Total: $ 90,702,811 Sim #: 2,000 (A: 123, B: 275, C: 213, D: 169) CAGR: 57.46 %
Total: $ 1,494,238,108 Sim #: 3,000 (A: 135, B: 292, C: 206, D: 147) CAGR: 89.80 %
Total: $ 266,142,475 Sim #: 4,000 (A: 123, B: 274, C: 232, D: 151) CAGR: 69.18 %
Total: $ 171,492,428 Sim #: 5,000 (A: 121, B: 279, C: 222, D: 158) CAGR: 64.29 %
Total: $ 1,826,281,096 Sim #: 6,000 (A: 133, B: 273, C: 242, D: 132) CAGR: 92.36 %
Total: $ 78,536,379 Sim #: 7,000 (A: 128, B: 261, C: 220, D: 171) CAGR: 55.96 %
Total: $ 1,156,130,503 Sim #: 8,000 (A: 149, B: 242, C: 244, D: 145) CAGR: 86.58 %
Total: $ 786,148,322 Sim #: 9,000 (A: 124, B: 283, C: 234, D: 139) CAGR: 81.84 %
Total: $ 497,823,126 Sim #:10,000 (A: 131, B: 278, C: 218, D: 153) CAGR: 76.39 %
```

Average : \$ 866,430,474 CAGR: 83.03 %

([Click here to enlarge](#))

Of the 10,000 simulations, we do have some under \$100 million, but most are above. And even higher than the estimated outcome of equation (5). Rerunning the simulation produced Figure #8 below. The numbers are similar to Figure #7.

### Figure #8: WL8 Simulation Using OPPW Portfolio Metrics - 2nd Sim.

```
# of Sims: 10,000 over 15 years using randomly selected trades based on WL8 sim metrics.
Total: $ 190,430,568 Sim #: 1 (A: 118, B: 276, C: 235, D: 151) CAGR: 65.44 %
Total: $ 310,740,480 Sim #: 1,000 (A: 129, B: 284, C: 206, D: 161) CAGR: 70.93 %
Total: $ 50,351,383 Sim #: 2,000 (A: 119, B: 272, C: 217, D: 172) CAGR: 51.40 %
Total: $ 714,230,248 Sim #: 3,000 (A: 128, B: 294, C: 207, D: 151) CAGR: 80.69 %
Total: $ 96,120,137 Sim #: 4,000 (A: 115, B: 292, C: 207, D: 166) CAGR: 58.07 %
Total: $ 318,209,674 Sim #: 5,000 (A: 135, B: 273, C: 209, D: 163) CAGR: 71.20 %
Total: $ 188,747,301 Sim #: 6,000 (A: 132, B: 253, C: 237, D: 158) CAGR: 65.35 %
Total: $ 475,953,039 Sim #: 7,000 (A: 140, B: 265, C: 216, D: 159) CAGR: 75.86 %
Total: $ 2,065,585,956 Sim #: 8,000 (A: 139, B: 284, C: 214, D: 143) CAGR: 93.94 %
Total: $ 1,124,541,773 Sim #: 9,000 (A: 129, B: 288, C: 221, D: 142) CAGR: 86.24 %
Total: $ 613,502,332 Sim #:10,000 (A: 129, B: 280, C: 223, D: 148) CAGR: 78.86 %
```

Average : \$ 892,489,095 CAGR: 83.39 %

([Click here to enlarge](#))

We do not know which of the 20,000 scenarios will prevail, even any of those. However, the expected average is, nonetheless, the most expected outcome.

The WL8 simulation results might be on the low side of estimates. One of the reasons for making that point is that all we have is one occurrence of the TQQQ trading strategy based on the above-mentioned trading rules.

The simulations should have provided a range of outcomes with means centered around the WL8 portfolio metrics of Figure #2. Yet, we got much more.

We have in Figures #7 and #8 cases where the outcome is below \$100 million, some even lower than the result obtained with the WL8 simulation using the OPPW trading strategy. However, most have been higher since even the average for the 10,000 simulations came in over \$800 million with an average CAGR at the 83% level.

Making the Monte Carlo simulation using randomly selected trade types should span a wide range of outcomes, as the above two figures demonstrate. We can observe in those two simulations<sup>5</sup> that the trade types navigate near their respective averages.

Relatively small changes to the portfolio metrics can have quite an impact on the overall portfolio results.

Still, we cannot determine the exact outcome over the following 15 years.

However, if those simulations, as presented above, are any indication, we should expect even higher results than those obtained in the WL8 OPPW trading strategy.

## **THE OVERFITTING, WHERE IS IT?**

From the start, I stated that there was no survivorship bias in the TQQQ trading strategy simply because none was available. No trading decision was based on future price data. We didn't use past market data except for the underlying notion of a long-standing upward trend on which the strategy is capitalizing. If the market is going up, we're in. It is more like: there is a market, we're in. There is no survivorship bias, but what about curve-fitting?

Usually, when you look at stock trading programs, they have outstanding simulated past results. We declare them over-fitted when they do not live up to their future expectations. The case is often easy to make; a simple walk-forward can make that case. For example, see my article: [A Walk-forward Example](#), where you will find a vivid example.

Here, with this strategy, the WL8 simulation gave outstanding results. However, since

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<sup>5</sup> We could do thousands of those simulations and get similar results

it is a single and known time series, we should expect that it represents an average outcome for the strategy. We could start looking at overfitting if its future performance level is lower than anticipated or estimated. But, it is not the case.

In those thousands of simulations based on the 15-year portfolio metrics, the outcome continued to rise, as illustrated in Figure #1. Recent articles made the same point. Figure #1 is the result of 200,000 simulations with an expanding time horizon that should all have tended to the portfolio metric averages obtained after the 15-year WL8 simulation of the OPPW strategy.

The WL8 simulation result was on the low side of estimates, not even what could be considered the average performance level for this strategy. The simulations relied on the distribution of randomly selected trade types. It did not use economic or fundamental market data but responded to price variance. It responded to how much TQQQ's price could wobble in a week.

Where could the overfitting come from?

You got into a trade every first trading day of the week. There is no room to overfit the number of weeks in those 15 years. We can apply the same reasoning to Friday's time-limited close. All trades closed on Friday's close had no other procedure to exit a trade; therefore, we could not optimize that procedure.

No matter how the TQQQ price series evolved, there were only 4 types of exits, and all trades would be executed at the same price for every WL8 simulation using the OPPW program.

However, using the trade type distribution to determine the trading results could vary widely due to the quasi-random redistribution of trade type boundaries, which we can see in Figures #7 and #8, where we have a wide range of possible outcomes. Some of those outcomes are even lower than the WL8 simulation results.

The WL8 result is below the expected average outcome of the randomly selected trade types.

Based on those simulation results, I make the case that the OPPW strategy's outcome is under-fitted. And 10,000 simulations might also be sufficient to illustrate that point. However, it will take a few years to confirm this.

Instead of waiting a few years, why not improve the strategy now?

Why not add some code to reduce the impact of Type-D trades?

You can see those coming your way. Trades are declared Type-D trades only on Friday's close, not before. They usually happen in declining markets, so we should

add trading procedures to avoid those trades or alleviate their impact.

It might sound trivial, but it is not. We could use equation (5) and change the number of Type-D trades by 10, 20, or 30% depending on our ability to find trading procedures that would avoid those Type-D trades without using overfitting methods.

For the 10, 20, and 30% reduction in Type-D trades, we would have:

$$0.50 \cdot 100,000 \cdot (1 + 0.085)^{205} \cdot (1 + 0.031)^{205} \cdot (1 - 0.0)^{185} \cdot (1 - 0.075)^{185 \cdot (1 - 0.10)} = 1,103,293,898$$

$$0.50 \cdot 100,000 \cdot (1 + 0.085)^{205} \cdot (1 + 0.031)^{205} \cdot (1 - 0.0)^{185} \cdot (1 - 0.075)^{185 \cdot (1 - 0.20)} = 4,667,337,010$$

$$0.50 \cdot 100,000 \cdot (1 + 0.085)^{205} \cdot (1 + 0.031)^{205} \cdot (1 - 0.0)^{185} \cdot (1 - 0.075)^{185 \cdot (1 - 0.30)} = 19,744,543,856$$

Eliminating 50% of those Type-D trades would generate:

$$0.50 \cdot 100,000 \cdot (1 + 0.085)^{205} \cdot (1 + 0.031)^{205} \cdot (1 - 0.0)^{185} \cdot (1 - 0.075)^{185 \cdot (1 - 0.50)} = 353,348,289,777$$

That should be our incentive to find ways to reduce Type-D trades. The overall reward appears worthwhile of our time. You can also do more. Yet again, an unexpected move could improve your outcome.

The next step is to decide what you can or intend to do.

## Related Papers and Articles:

The One Percent a Week TQQQ Trading Program: [Making Improvements](#), and [Part II](#)

[The One Percent Per Week TQQQ Trading Strategy: MY EQUATION](#)

[One Percent Per Week Strategy: Some Trading Habits](#)

[One Percent Per Week Strategy: Trade Distribution](#)

[THE TQQQ 3x-LEVERAGED SCENARIO](#)

[For Your Retirement, You Need To Win. It Is Not A Wish](#)

[You Can Make It Big, Real Big. If You Want](#)      *With Excel template.*

[Make Your First \\$50M Before You Retire](#)

[Your Trading Rules Matter](#)

[Gain Your Financial Freedom](#)      *Free book.*

[High Stock Portfolio Returns? Easy](#)

[Stock Trading Strategy Alpha Generation](#)

There Is Always A Better Retirement Fund: [Part I](#), and [Part II](#)

The One Percent a Week Stock Trading Program: [Part VII](#), and [Part VIII](#)

The One Percent a Week Stock Trading Program: [Part V](#), and [Part VI](#)

The One Percent a Week Stock Trading Program: [Part III](#), and [Part IV](#)

The One Percent a Week Stock Trading Program: [Part I](#), and [Part II](#)

The Long-Term Stock Trading Problem: [Part I](#), and [Part II](#)

[QQQ To The Rescue](#)

[Take the Money and Keep it – II](#)

[Use QQQ - Make the Money and Keep IT](#)